

Coordinated Optimizing Planning for AC-DC Distribution Networks with Quantitative Energy Storage and Converter Lifetime Assessment

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Abstract—The increasing penetration of distributed power sources in distribution networks has raised concerns about power consumption. Configuring energy storage (ES) devices becomes an effective planning strategy for enhancing the consumption of distributed photovoltaics. However, ES can only cooperate with limited consumption owing to site factors and capacity constraints. This paper presents a coordinated planning scheme for distribution networks, utilizing ES to balance time-scale fluctuations and constructing medium-voltage direct current (MVDC) lines between regions to mitigate spatial-scale fluctuations. Further, in such a coordinated distribution network planning problem, the equivalent annual cost model is commonly used for estimating the lifetime of ES devices and converters, but it exhibits limitations, particularly in the neglect of charge-discharge cycles and thermal cycling of devices. To address these issues, an equipment lifetime assessment model is established for planning the optimal number, capacity, and sites of ES devices and converters. Moreover, the coordinated optimal planning problem considering equipment lifetime to minimize the total cost is formulated. Various linearization methods are applied to transform the nonconvex nonlinear problem into a mixed-integer second-order cone programming problem. Finally, the case study demonstrates the effectiveness and economy of the proposed model.

Index Terms—Distribution network, lifetime model, energy storage, converter, coordinated optimal planning.

NOMENCLATURE

Abbreviations

PV	Photovoltaic.
ES	Energy storage.
SOC	State of charge.
VSC	Voltage source converter.
MISOCP	Mixed integer second-order cone programming.
DoD	Depth of discharge.

Sets

$\mathbf{E}$	Set of ESs.
$\mathbf{T}$	Set of hours.
$\mathbf{N}^{\text{VSC}}$	Set of VSCs.
$\mathbf{N}^{\text{Sub}}$	Set of substations.

$\mathbf{N}$	Set of AC nodes.
$\mathbf{L}$	Set of load nodes.
π	Set of parent nodes.
δ	Set of child nodes.

Parameters

C_e^{I}	Investment cost per energy capacity of ES.
C_p^{I}	Investment cost per power capacity of ES.
S_B	Base value of the network power.
$C_{\text{con}}^{\text{I}}$	Investment cost per capacity of VSC.
$C_{\text{sta}}^{\text{I}}$	Fixed investment cost of VSC station.
C_l^{I}	Investment cost per unit length of MVDC line.
Y	Operation cycle.
T_t^{A}	Ambient temperature at time t .
T	Time period, for 24 hours.
ΔT	Unit time, for 1 hour.
CL	Standard number of VSC thermal cycles.
M, K	Sufficiently large positive numbers.

Variables

$d_{e,t}$	DoD of ES e at time t .
$\varepsilon_{e,t}$	Equivalent lost lifetime of ES e at time t .
ε_e	Equivalent lost lifetime for n cycles of ES e .
E_e	Rated state of charge of ES e .
S_e	Rated capacity of ES e .
$P_{i,t}^{\text{SD}}/Q_{i,t}^{\text{SD}}$	Active/reactive power from AC side to VSC at node i at time t .
$P_{i,t}^{\text{loss}}$	Active power loss of VSC at node i at time t .
S_i^{S}	Capacity of the VSC at node i .
$P_{i,t}^{\text{S}}/Q_{i,t}^{\text{S}}$	Active/reactive power from node i to the VSC at time t .
$I_{i,t}^{\text{S}}$	Square of the current amplitude from node i to the VSC at time t .
$U_{i,t}/U_{j,t}$	Square of the voltage amplitude at node i/j at time t .
$P_{i,t}^{\text{Sub}}/Q_{i,t}^{\text{Sub}}$	Active/reactive power injected into node i by substation at time t .
$\text{SOC}_{e,t}$	The state of charge of ES e at time t .
$T_{i,t}^{\text{J}}$	Junction temperature of VSC at node i at time t .
$\Delta T_{i,t}^{\text{J}}$	Junction temperature difference of VSC at node i at time t .
$U_{i,t}^{\text{SD}}$	Square of the AC side voltage amplitude of VSC at node i at time t .
$P_{i,t}^{\text{DC}}/P_{j,t}^{\text{DC}}$	Power injected into the DC side of the VSC at node i/j at time t .
$I_{ij,t}^{\text{DC}}$	Square of the DC current amplitude between

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$U_{i,t}^{\text{DC}}/U_{j,t}^{\text{DC}}$	node i and j at time t . Square of the DC voltage of the VSC at node i and j at time t .
$P_{e,t}^{\text{C}}/Q_{e,t}^{\text{C}}$	Active/reactive charging power of ES e at time t .
$P_{e,t}^{\text{D}}/Q_{e,t}^{\text{D}}$	Active/reactive discharging power of ES e at time t .
$P_{i,t}^{\text{G}}/Q_{i,t}^{\text{G}}$	Active/reactive power injected by power source at node i at time t .
$P_{j,t}^{\text{L}}/Q_{j,t}^{\text{L}}$	Active/reactive power of the load flowing out of node j at time t .
$P_{ij,t}^{\text{B}}/Q_{ij,t}^{\text{B}}$	Active/reactive power flowing from node i to j on AC line.
$I_{ij,t}^{\text{B}}$	Square of the current amplitude from node i to j on AC line at time t .
l_{ij}	Length of MVDC line from node i to j .
$C_{\text{IESS},e}$	Primary investment cost of ES e .
$\Delta T_{\text{cycle},i}$	Thermal cycle of the VSC at node i for a day.
CL_i	Number of thermal cycles of the VSC at node i .
$C_{\text{Icon},i}$	Primary investment cost of VSC at node i .
x_{ij}	Binary variable for building MVDC line between node i and j .
x_e^{E}	Binary variable for installing ES.

I. INTRODUCTION

With increasing global demand for energy and environmental concerns, the development of renewable energy has gained significant attention [1]. During the period 2023–2025, the average annual solar photovoltaic (PV) capacity is expected to increase from 130 GW in the main case to 165 GW in the accelerated case, accounting for almost 60% of the total renewable energy expansion [2]. The increasing of large-scale distributed PV systems has raised concerns regarding consumption within distribution networks [3].

The literature has elucidated effective planning strategies for absorbing excess PV power through the configuration of energy storage (ES) systems. ES optimization planning was proposed for high-PV-penetration distribution networks to minimize the investment and operation costs of the distribution system in [4]. Reference [5] optimized the sizing of ES and PV generation for producers and consumers to minimize the total cost. However, ES devices cannot fully balance large-scale PV output power owing to their limited capacity, high costs, and site factors [6]. In addition, it is difficult to coordinate the PV output and load mismatch between different regions. With the development of power electronics technology, DC transmission benefits from its large transmission capacity and low losses, facilitating the formation of hybrid AC-DC network architectures in distribution networks [7]. The PV consumption capacity can be further increased by interregional energy transmission through medium-voltage direct current (MVDC) lines. Thus, the challenges of PV power consumption can be addressed through the coordinated planning of ES and MVDC lines in distribution networks. ES devices achieve peak shaving and valley filling at the temporal scale [8], and MVDC lines realize interregional energy exchange at the spatial scale [9].

In such coordinated planning schemes, ES devices and converters are critical components that constitute a significant

portion of the overall investment costs. The lifetime of these components is a pivotal factor in the planning process, which influences operational replacements, system investments, and planning rationality. In existing planning, the lifetime assessment of equipment is generally reflected by a given lifetime or an equivalent annual cost model. The equivalent annual cost converts investment expenditure into a uniform sequence equivalent annual cost, which can be calculated by determining the equipment lifetime and annual discount rate [10]. For example, [11] calculated the investment cost in the system using the equivalent annual cost model by providing the annual discount rate and lifetime of the ES devices and transmission lines. Reference [12] established the annualized capital cost based on the equivalent annual cost, where the lifetime of the system and interest rate are constants. In [13], the lifetime of ES devices in the equivalent annual cost model is typically defined as the serviceable period before a predefined minimum capacity is reached.

However, the state of the equipment varies under different operating conditions, and the lifetimes of ES devices and converters are directly related to the depth of discharge and thermal cycles in the actual process. Therefore, it is not feasible to use a uniform lifetime value based on engineering experience to accurately depict the lifetime of each piece of equipment in the planning problems. This may lead to different planning results, which in turn result in the inefficient use of equipment with short operational lifetimes, high failure rates, and high losses [14].

In some operational problems, the lifetime of equipment is usually characterized in detail to achieve goals such as energy management and economic dispatch. To evaluate ES lifetime, Reference [15] formulated an optimal scheduling problem for an ES to minimize the operational cost, considering the depreciation cost of the battery for different lifetime losses. ES lifetime estimation methods have been proposed and applied to wind-diesel hybrid systems and microgrid systems by considering the actual system operating conditions [16][17]. A model of battery and supercapacitor life degradation was established in [18], aimed at the optimal management of the ES lifetime of plug-in hybrid electric vehicles. In terms of converter lifetime modeling, the power device thermal load and lifetime estimation based on the loading and strength analysis of devices were considered in wind power converters to reduce the cost of wind power [19]. Reference [20] proposed an estimation method for the lifetime of insulated gate bipolar transistor modules of wind turbine power converter systems considering the ambient temperature. The impact of power converters and cell lifetimes on the economics of PV cell systems at different installation locations was analyzed in [21].

However, most of the equipment capacity in the aforementioned studies was provided. The lifetime model for the operational problem was based on the selected equipment. In the case of planning problems, the goal is to obtain the most economical solution for the configuration of the system equipment, where the capacity and location of the equipment are unknown. This distinguishes the planning problem that considers the equipment lifetime from the models for other

scenarios that consider lifetimes. Moreover, the increase in the variable dimensions and coupling between variables increases the difficulty of solving the problem. For instance, the depth of discharge (DoD) of an ES is obtained by dividing the state of charge (SOC) variation by the rated SOC; thus, the form of DoD is the division of two continuous variables, as the rated SOC is a decision variable. In addition, the numbers of ES devices and converters should be integer variables in the planning problem; therefore, a rounding function is needed to calculate the integer numbers by the lifetime cycle process. Furthermore, the number and capacity of the equipment in the cost are variables that need to be determined; thus, there is the problem of multiplying a rounded integer variable with a continuous variable. These constraints render the planning problem a complex nonconvex nonlinear program. Reference [22] sizes microgrid energy storage systems based on genetic algorithms (GA) with the objective of operating costs. Reference [23] uses an elitist nondominated sorting GA-II (NSGA-II) to determine the optimal sizing and allocation of battery energy storage systems in a distribution network considering the lifespan of the batteries. A hybrid tabu search/particle swarm optimization (TS/PSO) algorithm was used to solve the optimal planning of batteries in the distribution grid [24]. However, heuristic algorithms do not always find optimal solutions, and their effectiveness may be influenced by the characteristics of the problem itself and the parameter settings [25][26]. Therefore, in this study, the original nonconvex nonlinear problem is transformed into a mixed-integer second-order cone programming (MISOCP) problem by devising a series of equivalent or linearized methods that can be solved using an exact algorithm.

In summary, this study developed a model to evaluate the lifetimes of ES devices and converters based on their specific characteristics. The model calculates the charge and discharge cycles of the ES devices during operation as well as the thermal cycle profile of the converter. Furthermore, a coordinated optimization planning scheme for ES devices and MVDC lines considering equipment lifetime is proposed for a distribution network. Moreover, the nonconvex nonlinear planning model was transformed into an MISOCP problem using multiple linearization methods. The main contributions of this study are summarized as follows:

(1) Compared to the equivalent annual cost model in existing studies by giving a uniform lifetime value, this paper establishes an accurate equipment lifetime model by mathematically modeling the physical process of ES charge/discharge depth and the converter thermal cycles in the planning problem.

(2) A coordinated planning model with ES and MVDC lines is established to promote the integration of PV generation in both spatial and temporal scales. The planning model not only determines the location and capacity of the equipment but also takes into account the number of replacements during the planning period based on the accurate lifetime model.

(3) The original nonconvex nonlinear programming model containing complex constraints includes absolute values, bilinear terms, the embedded rounding problem, and the

coupling of absolute values with logistic constraints. A complete linearization scheme is designed to transform it into a solvable MISOCP problem.

The remainder of this paper is organized as follows: Section II introduces the lifetime assessment model of ES devices and converters. Section III describes the framework and mathematical model for coordinated planning considering equipment lifetime. Section IV presents the linearization and processing methods for the planning model. Section V presents multiple case studies and comparisons. Finally, Section VI concludes the study.

II. ENERGY STORAGE AND CONVERTER LIFETIME ASSESSMENT MODEL

A. ES lifetime model

As a short-lived device in the grid, frequent and high multiplier charging and discharging will significantly shorten the ES lifetime and affect the economy of the system. Therefore, it is necessary to analyze the impact of the charging and discharging power and depth of the ES on its lifetime and then establish a quantitative lifetime assessment model.

The equations for the existing planning scenarios using the equivalent annual cost approach [11]-[13] are as follows: The primary investment cost of the ES during planning cycle Y is expressed as

$$C_{\text{IESS},e} = Y \times \frac{r(1+r)^{y_{\text{ESS}}}}{(1+r)^{y_{\text{ESS}}} - 1} \times (C_e^I E_e + C_p^I S_e S_B), \forall e \in \mathbf{E} \quad (1)$$

where r denotes the annual discount rate, y_{ESS} is the service lifetime of an ES, which is generally determined directly based on engineering experience.

The ES lifetime is affected by several factors, such as temperature, current, and the number of charging and discharging cycles. These operational differences cause a difference in the ES lifetime. Therefore, planning for an ES must consider the impact of the operation mode on its lifetime, and make decisions on the number of units to be used during the planning cycle. The main factors are the depth and time of charging and discharging. Based on the cumulative damage model of the battery analyzed by the National Renewable Energy Laboratory (NREL), USA [27], each discharge process causes an irreversible loss until the end of the battery life.

Defining the DoD (Depth of discharge) of the ES e at moment t is given by

$$d_{e,t} = \frac{|SOC_{e,t} - SOC_{e,t-1}|}{E_e}, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (2)$$

The relationship between the equivalent lost lifetime and DoD can be expressed as a linearizable function:

$$\varepsilon_{e,t} = k_e d_{e,t} + b_e, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (3)$$

where k_e and b_e are the parameters for a certain type of ES e .

Assuming that the ES is cycled n times during operation, the equivalent lost lifetime is

$$\varepsilon_e = \sum_{t=1}^n \varepsilon_{e,t}, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (4)$$

When $\varepsilon_e=1$, the ES is considered obsolete and must be

replaced with a new one. The investment cost of the ES coupled with its lifetime in the operating cycle Y can be expressed as

$$C_{\text{IESS},e} = \lceil \varepsilon_e \rceil \times (C_e^I E_e + C_p^I S_e S_B), \forall e \in \mathbf{E} \quad (5)$$

where ε_e is the equivalent lost lifetime for n cycles of ES e . $\lceil \cdot \rceil$ denotes the rounding function and $\lceil \varepsilon_e \rceil$ denotes the total number of ES used during the operating cycle. C_e^I and C_p^I are the investment cost per energy capacity and per power capacity of ES. E_e is the rated state of charge of ES e . S_e is the rated capacity of ES e . S_B is the base value of the network power.

B. VSC lifetime model

The equation for calculating the primary investment cost of converters in the planning years using the equivalent annual cost in the existing planning [11]-[13] is given as

$$C_{\text{con},i} = Y \times \frac{r(1+r)^{y_{\text{con}}}}{(1+r)^{y_{\text{con}}} - 1} \times C_{\text{con}}^I S_i^S S_B + 2 \times C_{\text{sta}}^I x_{ij}, \forall i, j \in \mathbf{N}^{\text{VSC}} \quad (6)$$

Similar to (1), y_{con} is the service lifetime of the converter, which is generally determined directly based on engineering experience.

In this study, we establish a lifetime model that considers the operating characteristics of the converter and a cost model. The lifetime models provided by converter manufacturers are based on mathematical models that use data from aging tests. For example, in the lifetime model proposed by the ABB, the ratio of the thermal cycle number N_{life} to the failure rate B_{10} was used as a life index for power electronic equipment [28]. Herein, a voltage-source converter (VSC) lifetime assessment is performed by investigating the number of thermal cycles. A VSC consists of a converter bridge for AC/DC conversion. The VSC thermal cycle is primarily related to temperature changes in its components. The core component of a VSC is a semiconductor device, which is the main device that generates heat during operation. Therefore, the lifetime of a VSC depends on the junction temperature and its variation amplitude as well as the thermal cycle period [29]. Fig. 1 illustrates the proposed method for the VSC lifetime assessment in power networks.

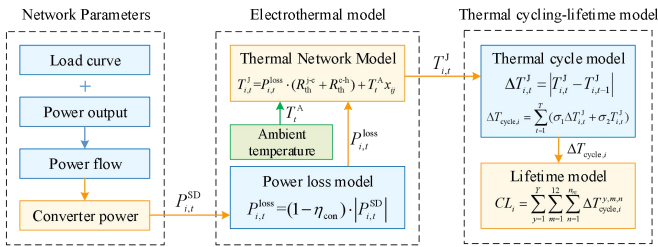


Fig. 1. Estimation model of VSC lifetime.

The first step is to calculate the power flow from the network parameters and obtain the power flowing through the converter. Next, based on the power-loss model, the active loss of the converter is obtained, and further combined with the current ambient temperature, the junction temperature of the converter is obtained using the electrothermal model. The junction temperature is then inputted into the thermal cycle model to obtain a single thermal cycle period. Finally, a single thermal cycle is accumulated to determine the number of thermal cycles

required to evaluate the lifetime of the converter. The actual lifetime of the converter is obtained by dividing the number of thermal cycles by the number of standard thermal cycles.

Practically, the main determinants of the VSC component temperature are variations in load power and ambient temperature. Power flow through the VSC generates power losses that heat the semiconductor device. The operation at different powers led to differences in the numbers of equivalent thermal cycles. For example, when the VSC operates at high power, the power loss, junction temperature, and equivalent thermal cycles increase; therefore, the lifetime loss increases. The ambient temperature affects VSC by affecting the heat dissipation process of the VSC components, thus affecting the thermal cycles and lifetime of the VSC.

The power loss of VSC can be expressed as:

$$P_{i,t}^{\text{loss}} = (1 - \eta_{\text{con}}) \cdot |P_{i,t}^{\text{SD}}|, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (7)$$

where η_{con} is the VSC efficiency. $P_{i,t}^{\text{SD}}$ is the active power from the AC side to the VSC at node i at time t .

As shown in Fig. 1, the VSC power loss is used as an input to the thermal network to calculate the VSC junction temperature.

$$T_{i,t}^j = P_{i,t}^{\text{loss}} \cdot (R_{\text{th}}^{j-c} + R_{\text{th}}^{c-h}) + T_t^A x_{ij}, \forall i, j \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (8)$$

where R_{th}^{j-c} is the thermal resistance from the junction to the case, R_{th}^{c-h} is the thermal resistance from the case to the heat sink, and T_t^A is the ambient temperature at time t . x_{ij} is a binary variable for building MVDC lines between nodes i and j . If built, the value is 1; otherwise, it is 0. Because the impact of short thermal cycles (< 1 s) on the VSC lifetime is small, the thermal capacitance inside the device and heat sink is ignored, and only the thermal resistance is considered in the long-term analysis.

The absolute value of the difference between the junction temperature at moments t and $t-1$ is:

$$\Delta T_{i,t}^j = |T_{i,t}^j - T_{i,t-1}^j|, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (9)$$

$$T_{i,T}^j = T_{i,0}^j, \forall i \in \mathbf{N}^{\text{VSC}} \quad (10)$$

The thermal cycle of the VSC for a day is obtained as the cumulative temperature difference.

$$\Delta T_{\text{cycle},i} = \sum_{t=1}^T (\sigma_1 \Delta T_{i,t}^j + \sigma_2 T_{i,t}^j), \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (11)$$

where σ_1 and σ_2 are scale factors.

Based on Miner's linear fatigue accumulation theory [30], the number of thermal cycles of the VSC at node i in Y years is extended to

$$CL_i = \sum_{y=1}^Y \sum_{m=1}^{12} \sum_{n=1}^{n_m} \Delta T_{\text{cycle},i}^{y,m,n}, \forall i \in \mathbf{N}^{\text{VSC}} \quad (12)$$

where y is the number of years, m is the number of months, and n_m is the number of days in the month m .

If the investment cost of the equipment remains constant, the average cost of the equipment over its operating cycle decreases as its lifetime increases. Assuming that the standard number of thermal cycles CL for each VSC is constant, the investment cost of the VSC and its station, coupled with its lifetime in operating cycle Y at node i , can be expressed as

$$C_{\text{con},i} = \left\lceil \frac{CL_i}{CL} \right\rceil \times C_{\text{con}}^I S_i^S S_B + 2 \times C_{\text{sta}}^I x_{ij}, \forall i, j \in \mathbf{N}^{\text{VSC}} \quad (13)$$

Here, $\lceil CL_i / CL \rceil$ denotes the total number of VSCs used during the operating cycle. Because a DC line has two VSC stations at each end, a binary indicator variable multiplied by two is required to represent the cost of the stations.

III. COORDINATED PLANNING MODEL FOR AC-DC DISTRIBUTION NETWORKS CONSIDERING EQUIPMENT LIFETIME

A. Coordinated framework considering equipment lifetime

The coordinated planning framework of the ESs and MVDC lines, considering equipment lifetime, is shown in Fig. 2. Two independent distribution network areas, A and B, are connected to the corresponding substations. However, owing to regional and policy factors, there may be a surplus of PV that can be loaded into Zone A and less PV in Zone B. This results in lower PV utilization in Zone A and more power purchases from the main grid in Zone B, making the entire distribution area less economical. By planning and building MVDC distribution lines between Zones A and B, the two areas are linked to form a hybrid AC/DC distribution system for mutual energy benefits. Furthermore, because the PV output cannot exactly match the load demand at every moment, it is necessary to consider the ES configuration to achieve peak and valley power reductions in the two areas. In addition, an accurate evaluation of the lifetimes of VSC and ES is critical for optimizing the economy of the system during long-term planning, as their operational modes are closely related to their lifetimes. Hence, considering the lifetime of equipment is crucial in the planning process. The coordinated planning model of ES and DC devices considering their lifetimes is developed in detail in subsequent sections.

B. AC-DC Distribution network model

The AC/DC distribution network consists of MVDC lines, ES and AC distribution network. The core formulas are listed below, and other general formulas are referred to Appendix A.

1) MVDC line

A biterminal MVDC line is constructed to transmit the excess power generated in the high-penetration PV area to achieve interregional PV consumption, as shown in Fig. 3. It consists of an MVDC line and VSC stations at both terminals, whereas the converter stations are considered a combination of ideal VSC, virtual nodes, and equivalent impedances.

The power flow relationship at both terminals of the equivalent impedance is expressed by (A.1) - (A.4). The transmission capacity constraint of a VSC station is expressed by (A.5). Equation (14) represents the VSC transmission power constraint. (15)-(17) are the DC-line power-flow constraints. (18) and (19) express the voltage constraints on the DC line.

$$\begin{cases} P_{i,t}^{DC} = \eta_{con} P_{i,t}^{SD}, P_{j,t}^{DC} = \frac{1}{\eta_{con}} P_{j,t}^{SD}, x_{i,t}^{Dir} = 0 \\ P_{i,t}^{DC} = \frac{1}{\eta_{con}} P_{i,t}^{SD}, P_{j,t}^{DC} = \eta_{con} P_{j,t}^{SD}, x_{i,t}^{Dir} = 1 \end{cases} \quad (14)$$

$$\forall i, j \in \mathbf{N}^{VSC}, \forall t \in \mathbf{T}$$

$$P_{i,t}^{DC} + P_{j,t}^{DC} - I_{ij,t}^{DC} R_{ij}^{DC} = 0, \forall i, j \in \mathbf{N}^{VSC}, \forall t \in \mathbf{T} \quad (15)$$

$$U_{i,t}^{DC} - U_{j,t}^{DC} - 2R_{ij}^{DC} P_{i,t}^{DC} + (R_{ij}^{DC})^2 I_{ij,t}^{DC} = 0, \quad (16)$$

$$\forall i, j \in \mathbf{N}^{VSC}, \forall t \in \mathbf{T}$$

$$(P_{i,t}^{DC})^2 = U_{i,t}^{DC} I_{ij,t}^{DC}, \forall i, j \in \mathbf{N}^{VSC}, \forall t \in \mathbf{T} \quad (17)$$

$$(V_{min}^{DC})^2 \leq U_{i,t}^{DC} \leq (V_{max}^{DC})^2, \forall i \in \mathbf{N}^{VSC}, \forall t \in \mathbf{T} \quad (18)$$

$$U_{i,t}^{SD} \leq U_{i,t}^{DC}, \forall i \in \mathbf{N}^{VSC}, \forall t \in \mathbf{T} \quad (19)$$

where $x_{i,t}^{Dir}$ denotes the power flow direction along the DC line at time t . The value is 0 when the power flows from node i to node j and 1 when the power flows in the reverse direction. R_{ij}^{DC} is the resistance of the MVDC line between node i and node j . V_{min}^{DC} and V_{max}^{DC} are the lower and upper limits of the DC voltage, respectively.

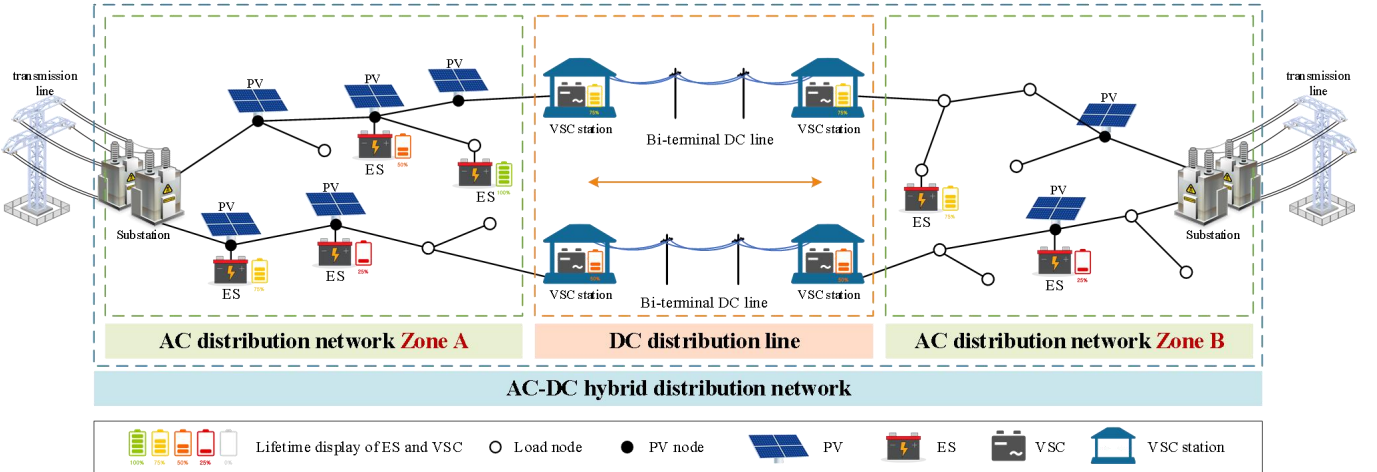


Fig. 2. Coordinated framework in AC-DC distribution networks considering ES and VSC lifetimes.

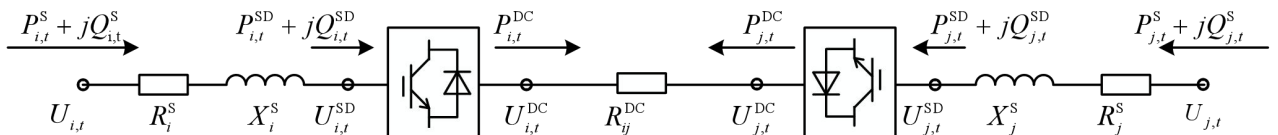


Fig. 3. Bi-terminal DC equivalent circuit.

The capacity planning constraint of the VSC station is expressed by (20):

$$0 \leq S_i^S \leq Mx_{ij}, \forall i, j \in \mathbf{N}^{\text{VSC}} \quad (20)$$

2) ES

The relationship between the ES siting decision and the charge/discharge state is shown in (21):

$$x_{e,t}^D + x_{e,t}^C \leq x_e^E, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (21)$$

where $x_{e,t}^D$ and $x_{e,t}^C$ are binary variables for the discharge/charge state of ES e at time t . x_e^E is the binary variable indicating whether ES e is placed in the predetermined position. If $x_e^E = 0$, both $x_{e,t}^D$ and $x_{e,t}^C$ are considered to be 0, that is there is no charging or discharging power at the location where the ES is not configured. If $x_e^E = 1$, simultaneous charging and discharging are not allowed at the location of the configured ES.

Considering that high-penetration PV may cause voltage overrun problems in the distribution network, the ES in this study can regulate reactive power, which can alleviate the problem and reduce losses. (A.6)-(A.9) are the constraints of the ES charging and discharging variables on power. (22) and (23) are ES capacity constraints.

$$(P_{e,t}^C)^2 + (Q_{e,t}^C)^2 \leq (S_e^C)^2, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (22)$$

$$(P_{e,t}^D)^2 + (Q_{e,t}^D)^2 \leq (S_e^D)^2, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (23)$$

The SOC of the ES is restricted to (24). The charging and discharging power constraints for the ES are expressed by (A.10) and (A.11).

$$0 \leq \text{SOC}_{e,t} \leq E_e, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (24)$$

The relationship between the ES siting decision and its capacity and charge is expressed in (25) and (26).

$$0 \leq S_e \leq Kx_e^E, \forall e \in \mathbf{E} \quad (25)$$

$$0 \leq E_e \leq Kx_e^E, \forall e \in \mathbf{E} \quad (26)$$

3) AC distribution network

The substation purchases power from the main grid to satisfy the regional power demand, which is treated as an equivalent generation node, as expressed in (27) and (28). Considering the possibilities of safety and acceptance problems when PV power is returned [31], the substation power is set not to be sent backwards, as shown in (29).

$$P_{i,t}^G = P_{i,t}^{\text{Sub}}, \forall i \in \mathbf{N}^{\text{Sub}}, \forall t \in \mathbf{T} \quad (27)$$

$$Q_{i,t}^G = Q_{i,t}^{\text{Sub}}, \forall i \in \mathbf{N}^{\text{Sub}}, \forall t \in \mathbf{T} \quad (28)$$

$$P_{i,t}^{\text{Sub}} \geq 0, \forall i \in \mathbf{N}^{\text{Sub}}, \forall t \in \mathbf{T} \quad (29)$$

Constraint (A.12) and (A.13) represent the power balance equation at node j . (A.14) and (A.15) show the relationships between the voltage and power flows at both terminals of the AC line. The line power constraint is expressed by (A.16) and the nodal AC voltage constraint is given by (A.17).

C. The objective function

1) Investment cost

The investment cost mainly consists of primary investment and construction costs. The primary investment cost includes the acquisition of the ES, MVDC line, and its two-end VSCs. Because ES often have a built-in matching converter, the cost of the converter for the ES is not considered separately. In addition, the scenario is a distribution system with installed PV;

therefore, the inverter for the PV is not included in the cost. The primary investment cost of MVDC lines is represented as

$$C_{\text{IL},ij} = C_1^l l_{ij} x_{ij}, \forall i, j \in \mathbf{N}^{\text{VSC}} \quad (30)$$

where C_1^l is the investment cost per unit length of MVDC line. l_{ij} is the length of MVDC line. x_{ij} is a binary variable for building MVDC line between node i and j .

The construction cost is generally selected as a percentage α of the primary investment cost [32]. According to (5), (13) and (30), the total investment cost including ES, VSC, VSC stations and MVDC lines are given as

$$C_1 = (1+\alpha)(C_{\text{IESS}} + C_{\text{lcon}} + C_{\text{IL}}) \\ = (1+\alpha) \left(\sum_{e \in \mathbf{E}} C_{\text{IESS},e} + \sum_{i \in \mathbf{N}^{\text{VSC}}} C_{\text{lcon},i} + \sum_{i,j \in \mathbf{N}^{\text{VSC}}} C_{\text{IL},ij} \right) \quad (31)$$

2) Maintenance cost

Maintenance costs include material and labor costs for equipment operation and maintenance. According to actual operation experience, the maintenance cost can be simplified to a certain percentage of the primary investment cost [32].

$$C_M = \beta_e C_{\text{IESS}} + \beta_{\text{con}} C_{\text{lcon}} + \beta_l C_{\text{IL}} \quad (32)$$

where β_e , β_{con} and β_l are the ratio of ES, VSC, and MVDC line maintenance costs to their primary investment costs, respectively.

3) Electricity cost

This study focuses on the long-term planning problem on an annual basis, it chooses one day as the smallest unit that can be stacked and calculated in a typical scenario. Because PV installations cannot fully satisfy the local load demand, power must be purchased from the public grid. Thus, the electricity cost is included, representing the cost of purchasing electricity from the main grid.

The electricity cost is the time of use electricity price multiplied by the power input from the main grid [33][34], and then extended to Y years:

$$C_p = Y \times 365 \sum_{\substack{i \in \mathbf{N}^{\text{Sub}} \\ t \in \mathbf{T}}} C_t^{\text{elec}} P_{i,t}^{\text{Sub}} S_B \Delta t \quad (33)$$

where C_t^{elec} is the electricity price at time t in a typical scenario and $P_{i,t}^{\text{Sub}}$ is the active power injected into node i by substation.

The purpose of planning in this study is to optimize and balance investment and operating costs to achieve the best overall system economics for high-penetration PV areas. By investing in ESs and MVDC lines, PV power can be absorbed as much as possible to reduce the cost of purchasing power from the main grid. Therefore, the objective function is the sum of the investment cost C_1 , maintenance cost C_M , and electricity cost C_p . The coordinated optimization model of the ES and MVDC lines, considering the lifetime in this study, is

$$\min C_1 + C_M + C_p \\ \text{s.t. (2)-(5),(7)-(33),(A.1)-(A.17)} \quad (34)$$

IV. LINEARIZATION AND PROCESSING METHODS

The above model describes the optimal planning of ESs and MVDC lines considering equipment lifetime. However, in this model, the power-flow relationship considering network loss involves a quadratic nonlinear expression. When calculating the ES lifetime, the absolute value of the difference in the SOC

is required to calculate the DoD. When calculating the VSC lifetime, the absolute values of the power and temperature differences must be calculated. To determine the power-flow efficiency of an MVDC line, a logical judgment statement must be introduced. An upward-rounding function is used to plan the amount of equipment required. The cost involves a nonlinear constraint on the multiplication of quantity and capacity. Overall, a series of nonlinear problems and possible numerical problems exist that exacerbate the difficulty in solving the problem. Therefore, this study applies relaxation and linearization methods to transform the problem into a standard MISOCP, thereby making the solution possible and rapid, as shown below:

A. Second-order cone relaxation

In Section III, (17), (A.4) and (A.15) are nonlinear and can be transformed into cone constraints via a second-order cone relaxation [35]. The detailed process is shown in Appendix B.

B. Polygon-based linearization

Nonlinear constraints, such as (A.5), (22) and (23) can be transformed using polygon-based linearization methods [36]. Let us consider (22) as an example.

$$-\sqrt{3}(P_{e,t}^c + S_e) \leq Q_{e,t}^c \leq -\sqrt{3}(P_{e,t}^c - S_e), \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (35)$$

$$-\frac{\sqrt{3}}{2}S_e \leq Q_{e,t}^c \leq \frac{\sqrt{3}}{2}S_e, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (36)$$

$$\sqrt{3}(P_{e,t}^c - S_e) \leq Q_{e,t}^c \leq \sqrt{3}(P_{e,t}^c + S_e), \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (37)$$

C. Absolute value

Constraints (2), (7), and (9) contain absolute-value functions, can be simplified by introducing binary variables. Let us consider (9) as an example:

$$\Delta T_{i,t}^j \geq T_{i,t}^j - T_{i,t-1}^j, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (38)$$

$$\Delta T_{i,t}^j \geq T_{i,t-1}^j - T_{i,t}^j, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (39)$$

$$\Delta T_{i,t}^j \leq T_{i,t-1}^j - T_{i,t}^j + (1 - x_i^{\text{Abs}})M, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (40)$$

$$\Delta T_{i,t}^j \leq T_{i,t}^j - T_{i,t-1}^j + x_i^{\text{Abs}}M, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (41)$$

where x_i^{Abs} are binary variables. If $T_{i,t}^j \geq T_{i,t-1}^j$, x_i^{Abs} is 1; otherwise, it is 0.

D. Logic constraints

Constraint (14) is a logical judgment constraint containing binary variables, which can be simplified into the following linear forms:

$$-Mx_{i,t}^{\text{Dir}} \leq P_{i,t}^{\text{SD}} \leq M(1 - x_{i,t}^{\text{Dir}}), \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (42)$$

$$-Mx_{i,t}^{\text{Dir}} \leq P_{i,t}^{\text{DC}} - \eta_{\text{con}} P_{i,t}^{\text{SD}} \leq Mx_{i,t}^{\text{Dir}}, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (43)$$

$$-M(1 - x_{i,t}^{\text{Dir}}) \leq P_{j,t}^{\text{DC}} - \eta_{\text{con}} P_{j,t}^{\text{SD}} \leq M(1 - x_{i,t}^{\text{Dir}}), \quad (44)$$

$$\forall i, j \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T}$$

$$-Mx_{i,t}^{\text{Dir}} \leq P_{j,t}^{\text{DC}} - \frac{1}{\eta_{\text{con}}} P_{j,t}^{\text{SD}} \leq Mx_{i,t}^{\text{Dir}}, \forall i, j \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (45)$$

$$-M(1 - x_{i,t}^{\text{Dir}}) \leq P_{i,t}^{\text{DC}} - \frac{1}{\eta_{\text{con}}} P_{i,t}^{\text{SD}} \leq M(1 - x_{i,t}^{\text{Dir}}), \quad (46)$$

$$\forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T}$$

E. Piecewise McCormick Envelope

Equation (2) in Section III is a nonlinear problem of dividing two variables, where both the SOC and capacity of the ES are continuous variables. The piecewise McCormick envelope is introduced and is mainly used to linearize the product of two bounded continuous variables [37]. The specific processing steps are as follows.

First, (2) can be transformed and simplified into a matrix form by multiplying the variables.

$$\mathbf{d} \in \mathbf{E} = \Delta \text{SOC} \quad (47)$$

where $\mathbf{d}$ and $\Delta \text{SOC} \in \mathbb{R}^{N_{\text{ES}} \times T}$. $\mathbf{E} \in \mathbb{R}^{N_{\text{ES}} \times T}$ is an expansion of E_e ; that is, $\mathbf{E} = E_e \cdot \text{ones}(1, T)$. N_{ES} is the number of ES.

Next, $\mathbf{d}$ and $\mathbf{E}$ are divided into N segments, defined as $\mathbf{d}^0 = [d_0, d_1, \dots, d_n, \dots, d_N]^T$ and $\mathbf{E}^0 = [E_0, E_1, \dots, E_n, \dots, E_N]^T$. $\mathbf{x}_d = [x_{d0}, x_{d1}, \dots, x_{dN}]^T$ and $\mathbf{x}_E = [x_{E0}, x_{E1}, \dots, x_{EN}]^T$. If $\mathbf{d}^0$ is in $[d_{n-1}, d_n]$, x_d is 1, otherwise 0. $\mathbf{x}_E$ follows the same rules as $\mathbf{E}$.

Thus, (2) can be expressed through McCormick segmental linearization as follows:

$$\begin{cases} \Delta \text{SOC} \geq \mathbf{E}^L \mathbf{d}^0 + \mathbf{d}^L \mathbf{E}^0 - (\mathbf{E}^L \mathbf{e} \mathbf{d}^L)^T \mathbf{g}_y \\ \Delta \text{SOC} \geq \mathbf{E}^U \mathbf{d}^0 + \mathbf{d}^U \mathbf{E}^0 - (\mathbf{E}^U \mathbf{e} \mathbf{d}^U)^T \mathbf{g}_y \\ \Delta \text{SOC} \leq \mathbf{E}^U \mathbf{d}^0 + \mathbf{d}^L \mathbf{E}^0 - (\mathbf{E}^U \mathbf{e} \mathbf{d}^L)^T \mathbf{g}_y \\ \Delta \text{SOC} \leq \mathbf{E}^L \mathbf{d}^0 + \mathbf{d}^U \mathbf{E}^0 - (\mathbf{E}^L \mathbf{e} \mathbf{d}^U)^T \mathbf{g}_y \end{cases} \quad (48)$$

$$\mathbf{x}_d \mathbf{e} \mathbf{d}^L \leq \mathbf{d}^0 \leq \mathbf{x}_d \mathbf{e} \mathbf{d}^U \quad (49)$$

$$\mathbf{x}_E \mathbf{e} \mathbf{E}^L \leq \mathbf{E}^0 \leq \mathbf{x}_E \mathbf{e} \mathbf{E}^U \quad (50)$$

where $\mathbf{x}_d$ and $\mathbf{x}_E$ are the binary variables. (51) ensures that the final $\mathbf{d}^0$ and $\mathbf{E}^0$ take values only at certain intervals. $\mathbf{y}$ denotes the product of $\mathbf{x}_d$ and $\mathbf{x}_E$, which is also a binary variable that can be simplified and expressed as (52).

$$1^T \mathbf{x}_E = 1, 1^T \mathbf{x}_d = 1 \quad (51)$$

$$\begin{cases} \mathbf{y} \leq \mathbf{x}_d \\ \mathbf{y} \leq \mathbf{x}_E \\ \mathbf{y} \geq \mathbf{x}_d + \mathbf{x}_E - 1 \end{cases} \quad (52)$$

Ultimately, the values of $\mathbf{d}$ and $\mathbf{E}$ are taken as:

$$\begin{cases} \mathbf{d} = 1^T \mathbf{d}^0 \\ \mathbf{E} = 1^T \mathbf{E}^0 \end{cases} \quad (53)$$

Thus, the expression for the piecewise McCormick is obtained.

F. Rounding process

Constraints (5) and (13) contain rounding functions, can be simplified by introducing integer and binary variables as follows. Let us consider (13) as an example.

The rounded value $\lceil CL_i / CL \rceil$ can be represented by an integer variable u_i^R ; the relationship between them should be

$$\frac{CL_i}{CL} \leq u_i^R < \frac{CL_i}{CL} + 1, \forall i \in \mathbf{N}^{\text{VSC}} \quad (54)$$

In practice, strict inequality has to be approximated as

$$\frac{CL_i}{CL} \leq u_i^R \leq \frac{CL_i}{CL} + 1 - \varepsilon, \forall i \in \mathbf{N}^{\text{VSC}} \quad (55)$$

where ε is a very small positive value.

Here, the integer variable u_i^R needs to be further simplified

because the integral values need to be multiplied by the continuous variables in (13). In binary form, the integer variable u_i^R is represented by three binary variables as follows:

$$u_i^R = 4x_i^{R0} + 2x_i^{R1} + x_i^{R2}, \forall i \in \mathbf{N}^{\text{VSC}} \quad (56)$$

Three binary variables are used because the number of ES or VSC required in the planning cycle is estimated to be no more than seven. If more variables are needed, the binary variables can also be expanded in this way.

$$C_{\text{con},i} = (4x_i^{R0} + 2x_i^{R1} + x_i^{R2}) \times C_{\text{con}}^I S_B + 2 \times C_{\text{sta}}^I x_{ij}, \quad \forall i, j \in \mathbf{N}^{\text{VSC}} \quad (57)$$

where u_i^R are integer variables that can be combined with binary integer variables for a binary representation. $u_i^R \in \{0, 1, \dots, 7\}$ denotes the number of VSC in cycle Y .

Furthermore, variables $y_i^{R0}, y_i^{R1}, y_i^{R2}$ are introduced to simplify the product of the binary and continuous variables in (57).

$$\begin{cases} -M(1-x_i^{R0}) \leq y_i^{R0} - 4S_i^S \leq M(1-x_i^{R0}) \\ -Mx_i^{R0} \leq y_i^{R0} \leq Mx_i^{R0} \end{cases}, \forall i \in \mathbf{N}^{\text{VSC}} \quad (58)$$

$$\begin{cases} -M(1-x_i^{R1}) \leq y_i^{R1} - 2S_i^S \leq M(1-x_i^{R1}) \\ -Mx_i^{R1} \leq y_i^{R1} \leq Mx_i^{R1} \end{cases}, \forall i \in \mathbf{N}^{\text{VSC}} \quad (59)$$

$$\begin{cases} -M(1-x_i^{R2}) \leq y_i^{R2} - S_i^S \leq M(1-x_i^{R2}) \\ -Mx_i^{R2} \leq y_i^{R2} \leq Mx_i^{R2} \end{cases}, \forall i \in \mathbf{N}^{\text{VSC}} \quad (60)$$

Thus, a simplification of the rounding is obtained using (61).

$$C_{\text{con},i} = (4y_i^{R0} + 2y_i^{R1} + y_i^{R2}) \times C_{\text{con}}^I S_B + 2 \times C_{\text{sta}}^I x_{ij}, \quad \forall i, j \in \mathbf{N}^{\text{VSC}} \quad (61)$$

G. Numerical issue

The model in this study is a long-term planning problem extended from one day to Y years, and there exists a superposition of small numbers to form large numbers, which may cause numerical problems in the program. For example, the loss lifetime of a single cycle of ES is small (10^{-4} – 10^{-3}), whereas the cost accumulates annually (10^5 – 10^9), leading to low accuracy and slow solving. Therefore, in this study, a program was written by adjusting the magnitude and merging variables to narrow the range of values and reduce their instability.

(1) Adjust the magnitude: The objective function is the total cost of long-term planning. If the operating cycle is 10 years, the magnitude of each type of cost is approximately 10^6 – 10^8 yuan. In this study, the program modification unit cost is thousand yuan, narrowing the scale to 10^3 – 10^5 and effectively solving the problem of a large number.

(2) Merge variables. In this program, (3) and (4) are combined to obtain the accumulated lost ES lifetime. The cumulant is used for the subsequent calculations to avoid the problem of small numbers.

Thus far, nonlinear relationships have been transformed into solvable constraints using various linearization methods. Other constraints similar to the aforementioned constraint treatment are linearized, as shown in Appendix. Thus, the original nonlinear nonconvex problem is transformed into an MISOCP problem as follows:

$$\min C_I + C_M + C_P$$

$$\text{s.t. } (3),(4),(8),(10)-(12),(15),(16),(18)-(21),(24)-(29), \quad (62) \\ (30)-(33),(35)-(61),(A.1)-(A.3),(A.6)-(A.14), \\ (A.16),(A.17),(B.2),(B.4),(B.6)-(B.31)$$

V. CASE STUDY

In this study, we proposed the use of MATLAB R2021b and Gurobi 10.0.0 for solving the problem. The distribution scenario in this study is divided into two areas (Zones A and B), each composed of a modified IEEE 13-node system. Therefore, the distribution network contained 26 nodes, including 24 load nodes. The topology is illustrated in Fig. 4.

Zones A and B are set up with multiple PV nodes. The PV in zone A has an overflow problem, and its maximum output is approximately 1.5 times the maximum load. No PV overflow problem occurred in zone B. PV output and daily load curves are shown in Fig. 5. Ambient temperature and electricity price curves are given in Fig. 6. Table I illustrates the parameters including AC/DC line parameters, unit cost of multiple items [32], lifetime of equipment [38], equivalent impedance, ES efficiency [39], thermal resistance, and standard thermal cycle number of the VSC [28] and so on. The ESs and MVDC lines are pre-sited before planning. Specifically, (i) ES is pre-sited at 24 load nodes, and (ii) MVDC lines are pre-sited between nodes 645–733, 646–734, 684–792, and 652–775, considering inter-regional distances and topographical reasons.

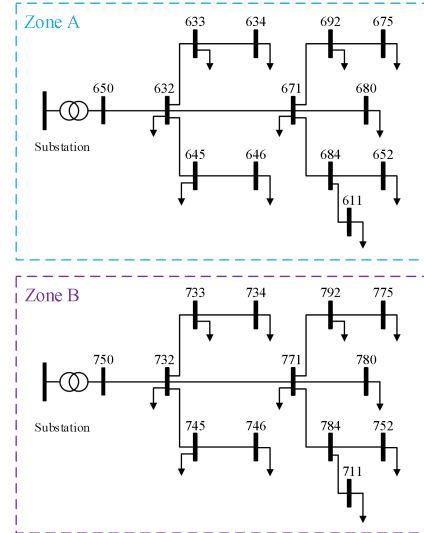


Fig. 4. Topology of 26-node distribution network.

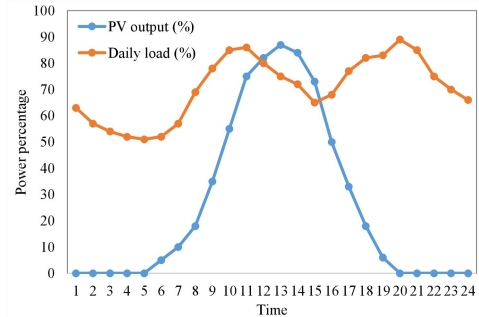


Fig. 5. Curves of PV output and load demand in a day.

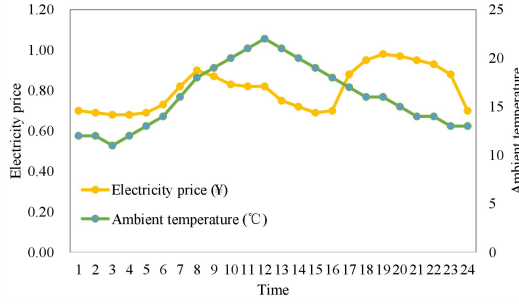


Fig. 6. Curves of ambient temperature and electricity price in a day.

TABLE I
SYSTEM PARAMETERS.

Parameter	Value
Resistance per unit length of AC line (Ω/km)	0.21
Reactance per unit length of AC line (Ω/km)	0.4
Resistance per unit length of DC line (Ω/km)	0.05
Equivalent resistance of VSC (Ω)	0.5
Equivalent reactance of VSC (Ω)	1.5
PV power factor $\cos\phi$	0.98
Efficiency of VSC	0.9
Efficiency of ES charging and discharging	0.95
Scale factor σ_1 in the VSC thermal cycle equation	0.8
Scale factor σ_2 in the VSC thermal cycle equation	0.2
Standard number of thermal cycles of VSC(k)	1000
Thermal resistance of VSC ($^{\circ}\text{C}/\text{MW}$)	500
Investment cost per unit SOC of ES ($\text{¥}/\text{kWh}$)	600
Investment cost per unit capacity of ES ($\text{¥}/\text{kVA}$)	1000
Investment cost per unit capacity of VSC ($\text{¥}/\text{kVA}$)	300
VSC station costs (k ¥)	20
Investment cost per unit length of DC line ($\text{¥}/\text{m}$)	40
Ratio of maintenance cost to primary investment cost β	0.02
Ratio of construction cost to primary investment cost α	0.05
Equal annual value discount rate	0.06
Equivalent annual value service lifetime of VSC (year)	20
Equivalent annual value service lifetime of ES (year)	5
Equivalent annual value service lifetime of DC line (year)	30
Equivalent annual value service lifetime of VSC station (year)	30
Resistance per unit length of AC line (Ω/km)	0.21
Reactance per unit length of AC line (Ω/km)	0.4

With the objective of minimizing the total cost in the operating cycle Y (set as 10 years), the coordinated planning results of the ES and MVDC lines considering the lifetime are solved, as shown in Fig. 7. The solution was obtained for the configuration of four ESs at nodes 671, 652, 732, and 745, and the DC distribution lines in the 684–792 and 652–775 node intervals. Coordinated optimization connects the original independent A and B distribution areas using MVDC lines.

Furthermore, to compare the characteristics and differences in the results under different models, another planning model based on the equivalent annual cost is proposed and implemented. Results for the same scenario are obtained and compared with those of the proposed model. For this purpose, four cases are proposed to compare whether the equipment lifetime is considered (the lifetime model in this study or the equivalent annual cost model) and whether the optimization is coordinated (configuring ES and MVDC lines, or only ES).

Case 1 is a coordinated planning considering the lifetimes of the ES and VSC (the model in this study).

Case 2 involved coordinated planning using the equivalent annual cost method (model in [11]–[13]).

Case 3 is the planning with only ES configuration

considering ES lifetime.

Case 4 is the planning with only ES configuration using the equivalent annual cost (model in [4]).

Subsequently, the four cases are analyzed and compared in terms of equipment lifetime, coordinated planning strategy, and economic costs.

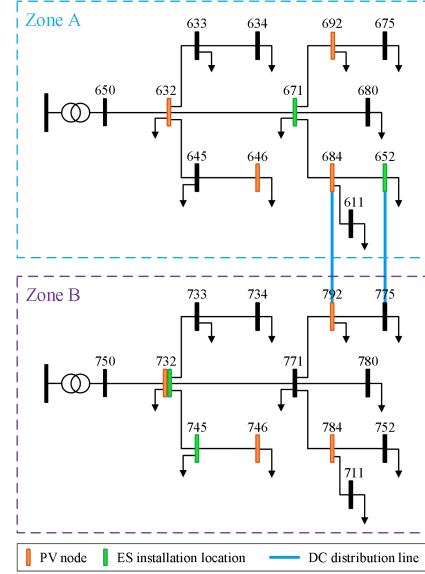


Fig. 7. The coordinated planning results of 26-node distribution network.

A. Equipment lifetime comparison

This section analyzes the effect of equipment lifetime on the results, focusing on comparing the differences between the proposed lifetime model and equivalent annual cost model. Cases 1 and 2 involved a coordinated planning. Case 1 is based on the proposed lifetime model, and the variables related to the ES and VSC lifetimes are to be optimized. Case 2 is based on the equivalent annual cost model, where the lifetime is an undifferentiated parameter based on engineering experience. Cases 3 and 4 are both ES-only plans, and the difference between them is the lifetime model. Table II and Table III summarize the specific configuration results for the ES and MVDC lines in the four cases.

TABLE II
CONFIGURATION RESULTS OF ES FOR FOUR CASES.

Case	Num.	Position	Capacity (MVA)	Max SOC (MWh)	Lifetime (year)	Average lifetime (year)
1	4	671	0.0750	0.2586	11.60	12.65
		652	0.0750	0.3103	10.00	
		732	0.0750	0.1552	14.50	
		745	0.0750	0.1552	14.50	
2	2	671	0.0750	0.2094	2.63	2.63
		711	0.0750	0.0843	2.63	
		632	0.0836	0.3000	10.00	
		633	0.0750	0.3103	10.00	
		645	0.0750	0.3103	10.00	
		671	0.0750	0.3103	10.00	
		692	0.0750	0.3103	10.00	
		675	0.0750	0.3103	10.00	
3	11	611	0.1197	0.5690	5.27	9.57
		652	0.0839	0.3013	10.00	
		680	0.0750	0.3013	10.00	
		745	0.1294	0.2586	10.00	
		771	0.2314	0.2988	10.00	
		645	0.1239	0.4707	2.63	
		633	0.0750	0.3103	10.00	
		671	0.0750	0.3103	10.00	
4	7	645	0.1239	0.4707	2.63	2.71

646	0.0750	0.2368	2.63
671	0.1365	0.4620	2.63
675	0.1132	0.4122	3.06
611	0.0899	0.2852	2.72
771	0.2318	0.1921	2.63
711	0.1273	0.1006	2.63

TABLE III
CONFIGURATION RESULTS OF MVDC LINES FOR FOUR CASES.

Case	DC line position	Length	VSC capacity (kVA)	Lifetime (year)		Average
				at zone A	at zone B	
1	684-792	609.6	225.2	24.64	25.33	25.06
	652-775	609.6	230.4	24.77	25.51	
2	646-734	609.6	257.8	25.03	25.76	24.61
	684-792	609.6	344.9	23.32	24.34	
3	-	-	-	-	-	-
4	-	-	-	-	-	-

Note: Symbol “-” denotes that this item doesn’t exist.

According to Table II and Table III, the average lifetime of the ES configured in Case 1 is approximately 12.65 y, whereas that of the VSC is 25.06 y. The average lifetime of the ES configured in Case 2 is 2.63 y, with a VSC of 24.61 y. The average lifetimes of the ES in Cases 3 and 4 are 9.57 y and 2.71 y, respectively. Therefore, the ES lifetime is significantly improved by considering the lifetime assessment model in the optimization. The lifetime of VSC is not significantly improved, but its average capacity is reduced from 301.4 to 227.8 kVA with a 24.4% reduction, which also illustrates the economic improvement because of the consideration of lifetime. In addition, ES devices are configured at more nodes and with smaller individual capacities in Cases 1 and 3 (with 4 and 11 ESs) compared to Cases 2 and 4 (with 2 and 7 ESs). So, it can be seen that the optimal result considering lifetime is to allocate small and decentralized ESs.

Multiple power flow curves are shown in Fig. 8 to study the operation based on the optimization results.

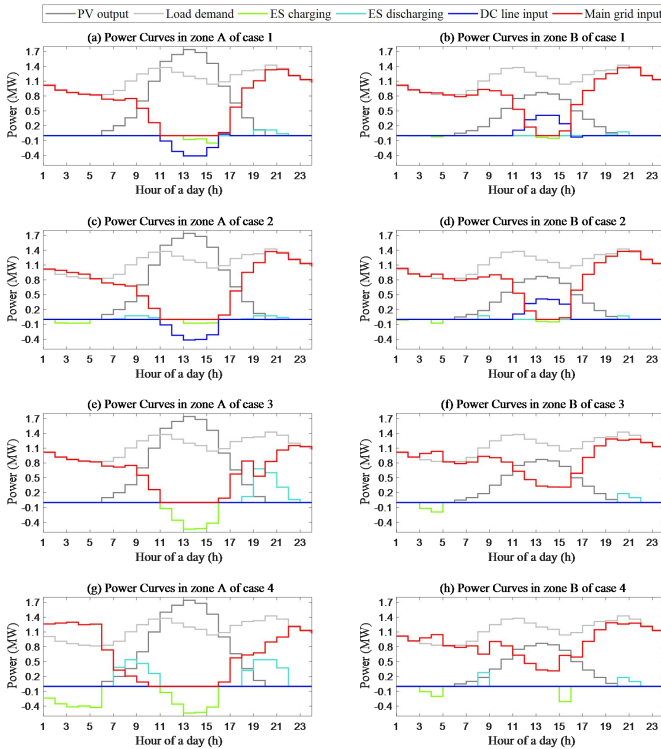


Fig. 8. Multiple power curves in zone A and B of four cases.

As shown in Figs. 8 (a) and (e), the operation of the ES tends to be conservative after considering the optimized lifetime; that is, it will hardly be charged in the early morning and will not be discharged in the morning. By contrast, under the equivalent annual cost model, as shown in Figs. 8 (c) and (g), the ES is charged and discharged frequently, which logically causes a decrease in the lifetime.

In addition, by dividing the SOC of the ES by the rated SOC, the DoD of the ES at each moment during the operation is obtained. The DoD range of the ES during the day is depicted in Fig. 9.

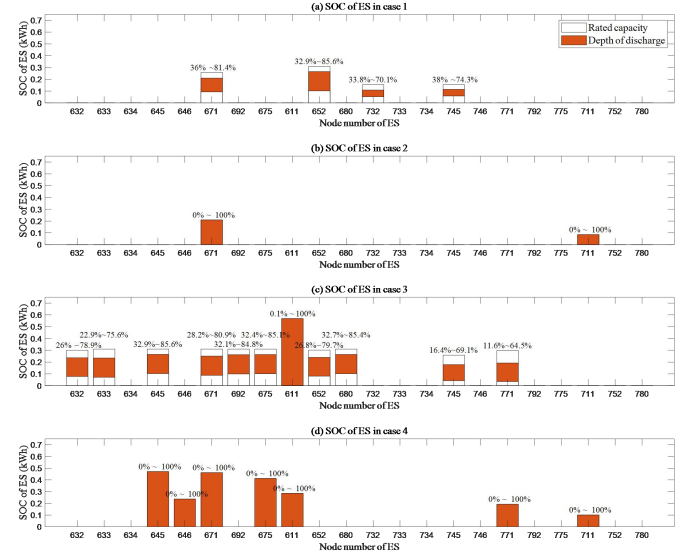


Fig. 9. SOC of ES in four cases.

As can be seen from Fig. 9, the DoD of each ES configured in Case 1 is maintained at approximately 30%–85%, whereas the ES in Cases 2 and 4 is fully charged and discharged within a range of 0%–100%. The results indicate that maintaining the charging and discharging interval of ES at around 30–85% is a better method, which can reduce lifetime loss by avoiding deep usage. In optimal planning, not only the rated SOC but also the SOC at each moment is the variable to be optimized. When the basic requirements are met, a smaller SOC results in a larger DoD, causing more energy loss and a shorter lifetime, and hence requiring more frequent replacement of the ES during operation. Conversely, a larger SOC leads to a smaller DoD but also results in a higher investment cost. Therefore, it is important to achieve an effective compromise between the SOC and DoD of the ES. A common practice that does not consider the ES lifetime is to maintain the DoD at approximately 0%–100%. This is a full charge and discharge cycle, but it can significantly reduce the lifetime of the ES and may not be optimal for planning systems designed for long ES lifetimes. By optimizing the DoD, it is possible to strike a balance between minimizing costs and extending the lifetime of the ES. Furthermore, the DoD for nearly all ES remains within the range of 10%–85% in Case 3. However, the SOC for the ES at Node 611 is significantly higher and reaches full discharge. This discrepancy can be attributed to the location of node 611, which is located at the end of the line and experiences a heavy load. In the absence of an MVDC line to enable interregional

energy sharing, the power demand for Node 611 requires the use of a short-lived, high-capacity ES, which is economically feasible. Therefore, economic results considering the equipment lifetime are obtained according to scenario changes, and the goal is to achieve an optimal system economy.

B. Coordinated planning strategy analysis

This section analyzes the effect of co-optimization on the results, focusing on building MVDC lines. From Table II, Case 1 is configured with four ESs and two MVDC lines, Case 2 is configured with two ESs and two MVDC lines, Case 3 with eleven ESs, and Case 4 with seven ESs. Cases 1 and 2 with MVDC lines have significantly smaller ES. This is because the lines mainly account for the power consumption and transmission of the PV overflow during high generation, partially replacing ES charging.

The main contribution of the MVDC line is the transmission of excess power to Zone B when the PV overflows in Zone A, thus reducing the power purchased by Zone B from the main grid at midday, as shown in Figs. 8 (a), (b), (c), and (d). This not only achieves the consumption of PV in Zone A but also enhances the economy of Zone B. As for the ES, the main role is to cut power peaks and fill valleys, as well as to consume PV in situ, specifically: (1) charging during high PV generation in zone A and discharging in the evening when there is low PV generation, high demand, and

high electricity prices, as shown in (a), (c), (e), and (g) of Fig. 8. For example, in Case 3, the ESs charge at 11–16 pm and discharging at 18–23 pm. (2) Power is purchased from the main grid for charging in the early morning and midday when prices are low and discharged during the morning or evening when demand is high, as shown in Figs. 8 (b), (c), (d), (f), (g), and (h). For example, in Case 4, the ESs charge at 0–6 am and discharging at 7–11 am and 18–22 am. Overall, the result of coordinated planning is that MVDC lines and ESs have their respective characteristics, which link two independent areas to form a hybrid AC-DC distribution system, enhancing the economy of the entire system.

C. Economic cost comparison

This section compares and analyzes the differences between the multiple costs for each case. The total cost during the operation cycle (10 y) is divided into three parts: investment, maintenance, and electricity costs. The multiple costs of the four cases are listed in Table IV. As Table IV illustrates, Case 1 has the lowest total cost, which is due to the low investment and maintenance costs of the equipment. Conversely, Case 2 has four replacements over 10 y because of the low lifetime of the ES; therefore, the investment cost is increased, resulting in a higher total cost. Although Cases 3 and 4 have low electricity costs, they have a large ES (Case 3) and a low lifetime (Case 4), which also results in a poor economy.

TABLE IV
COMPARISON OF MULTIPLE COSTS FOR FOUR CASES.

Case	ES	Investment Cost (M¥)		MVDC line	Maintenance cost (M¥)	Electricity cost (M¥)	Total cost (M¥)
		VSC	VSC station				
1	0.869	0.287	0.084	0.051	0.025	109.315	110.631
2	1.370	0.380	0.084	0.051	0.036	109.357	111.278
3	3.899	-	-	-	0.074	107.319	111.292
4	9.212	-	-	-	0.176	106.706	116.093

Note: Symbol “-” denotes that this item doesn’t exist.

In summary, Case 1 presented herein is an economically optimal solution with the objective of minimizing the total cost. The idea is to simultaneously plan the ESs and MVDC lines, considering the ES and VSC lifetimes. It can effectively absorb PV power and perform interregional energy mutualization through energy transfer to reduce electricity costs. Compared with the system without MVDC lines construction (Case 3), Case 1 can significantly reduce the power purchase cost in the PV deficient area, resulting in a further total cost of ¥661,000. Simultaneously, by optimizing the lifetime, the operation strategy is adjusted to effectively improve the lifetime of the ES and VSC to reduce the number of replacements in the cycle and further reduce the investment cost. Compared with Case 2, which uses the equivalent annual cost to simulate the lifetime, Case 1 can reduce the total cost by ¥647,000.

VI. CONCLUSION

A coordinated optimal planning model for ES and MVDC lines considering equipment lifetime is proposed in this study. Considering the temporal and spatial mismatch between the PV power output and load demand, an active distribution network

architecture with ES and MVDC lines is established. Furthermore, the lifetimes of the ES and VSC are considered a constraint for medium- and long-term economic planning in a hybrid AC-DC distribution network. To solve this nonlinear optimization problem, a series of processing schemes are designed to form an MISOPC problem.

The results show that the planning scheme can effectively achieve peak and valley reductions at different times through ES, and improve the supply and demand imbalance between regions through MVDC lines. It is worth mentioning that the results show that the charging and discharging interval of the ES is maintained at approximately 30%–85%, which indicates that the model considering the lifetime issue can reduce the lifetime loss by avoiding deep usage. Compared with the system without MVDC construction, this system can significantly reduce the power purchase cost in the PV deficient area by a further total cost of ¥661,000. Compared with the classical scheme, which uses the equivalent annual cost to simulate the equipment lifetime, this scheme has a more theoretical basis for the lifetime calculation, considering that the lifetime in the planning problem extends the ES lifetime

from 2.63 to 12.65 y on average and reduces the average planning capacity of the VSC by 24.4%, thus reducing the total cost by ¥647,000. In summary, our proposed method, which considers the lifetime of ES devices and MVDC lines, can provide a more economical and effective planning scheme in a distribution network. 公式章 (下一章) 节 1

APPENDIX

A. AC-DC Distribution network

1) MVDC line

$$P_{i,t}^S - I_{i,t}^S R_i^S = P_{i,t}^{SD}, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (\text{A.1})$$

$$Q_{i,t}^S - I_{i,t}^S X_i^S = Q_{i,t}^{SD}, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (\text{A.2})$$

$$U_{i,t} - U_{i,t}^{SD} - 2(R_i^S P_{i,t}^S + X_i^S Q_{i,t}^S) + [(R_i^S)^2 + (X_i^S)^2] I_{i,t}^S = 0, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (\text{A.3})$$

$$I_{i,t}^S = \frac{(P_{i,t}^S)^2 + (Q_{i,t}^S)^2}{U_{i,t}}, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (\text{A.4})$$

$$(P_{i,t}^{SD})^2 + (Q_{i,t}^{SD})^2 \leq (S_i^S)^2, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (\text{A.5})$$

where R_i^S and X_i^S denote the equivalent resistance and reactance of the VSC station at node i , respectively.

2) ES

$$0 \leq P_{e,t}^D \leq Mx_{e,t}^D, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (\text{A.6})$$

$$0 \leq P_{e,t}^C \leq Mx_{e,t}^C, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (\text{A.7})$$

$$0 \leq Q_{e,t}^D \leq Mx_{e,t}^D, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (\text{A.8})$$

$$0 \leq Q_{e,t}^C \leq Mx_{e,t}^C, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (\text{A.9})$$

$$SOC_{e,t} = SOC_{e,t-1} + (\eta_e^C P_{e,t-1}^C - \frac{1}{\eta_e^D} P_{e,t-1}^D) \Delta T, \quad \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (\text{A.10})$$

$$SOC_{e,T} = SOC_{e,0}, \forall e \in \mathbf{E} \quad (\text{A.11})$$

where η_e^C and η_e^D are the charging and discharging efficiencies, respectively.

3) AC distribution network

$$P_{j,t}^L \leq P_{j,t}^D - P_{j,t}^C + P_{j,t}^G + \sum_{i \in \pi(j)} (P_{ij,t}^B - I_{ij,t}^B R_{ij}) - \sum_{k \in \delta(j)} P_{jk,t}^B - \sum_{j \in \mathbf{N}^{\text{VSC}}} P_{j,t}^S, \forall j \in \mathbf{N}, \forall t \in \mathbf{T}, \quad (\text{A.12})$$

$$Q_{j,t}^L \leq Q_{j,t}^D - Q_{j,t}^C + Q_{j,t}^G + \sum_{i \in \pi(j)} (Q_{ij,t}^B - I_{ij,t}^B X_{ij}) - \sum_{k \in \delta(j)} Q_{jk,t}^B - \sum_{j \in \mathbf{N}^{\text{VSC}}} Q_{j,t}^S, \forall j \in \mathbf{N}, \forall t \in \mathbf{T}, \quad (\text{A.13})$$

$$U_{i,t} - U_{j,t} - 2(R_{ij} P_{ij,t}^B + X_{ij} Q_{ij,t}^B) + (R_{ij}^2 + X_{ij}^2) I_{ij,t}^B = 0, \quad \forall ij \in \mathbf{L}, \forall t \in \mathbf{T}, \quad (\text{A.14})$$

$$I_{ij,t}^B = \frac{(P_{ij,t}^B)^2 + (Q_{ij,t}^B)^2}{U_{i,t}}, \forall ij \in \mathbf{L}, \forall t \in \mathbf{T} \quad (\text{A.15})$$

where R_{ij} and X_{ij} are the resistance and reactance of AC line, respectively.

$$(P_{ij,t}^B)^2 + (Q_{ij,t}^B)^2 \leq (S_{ij,\max}^B)^2, \forall ij \in \mathbf{L}, \forall t \in \mathbf{T} \quad (\text{A.16})$$

$$(V_{\min}^{\text{AC}})^2 \leq U_{i,t} \leq (V_{\max}^{\text{AC}})^2, \forall i \in \mathbf{N}, \forall t \in \mathbf{T} \quad (\text{A.17})$$

where $S_{\max}^{\text{B},ij}$ is the maximum transmission capacity of the AC line. $V_{\min}^{\text{AC}}$ and $V_{\max}^{\text{AC}}$ are the lower and upper limits, respectively, of a given AC voltage.

B. 公式节 (下一节) Linearization and processing methods

1) Second-order cone relaxation

(A.4) can be relaxed to an inequality form, as (B.1), which does not affect the optimal solution. It can then be further organized into a standard second-order cone form, as (B.2).

$$I_{i,t}^S \geq \frac{(P_{i,t}^S)^2 + (Q_{i,t}^S)^2}{U_{i,t}}, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (\text{B.1})$$

$$\left\| \begin{array}{c} 2P_{i,t}^S \\ 2Q_{i,t}^S \\ I_{i,t}^S - U_{i,t} \end{array} \right\|_2 \leq I_{i,t}^S + U_{i,t}, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (\text{B.2})$$

(17) is transformed into (B.3) and further into (B.4).

$$(P_{i,t}^{\text{DC}})^2 \leq U_{i,t}^{\text{DC}} I_{ij,t}^{\text{DC}}, \forall i, j \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (\text{B.3})$$

$$\left\| \begin{array}{c} 2P_{ij,t}^{\text{DC}} \\ I_{ij,t}^{\text{DC}} - U_{i,t}^{\text{DC}} \end{array} \right\|_2 \leq I_{ij,t}^{\text{DC}} + U_{i,t}^{\text{DC}}, \forall i, j \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (\text{B.4})$$

(A.15) is transformed into (B.5) and further into (B.6).

$$I_{ij,t}^B \geq \frac{(P_{ij,t}^B)^2 + (Q_{ij,t}^B)^2}{U_{i,t}}, \forall ij \in \mathbf{L}, \forall t \in \mathbf{T} \quad (\text{B.5})$$

$$\left\| \begin{array}{c} 2P_{ij,t}^B \\ 2Q_{ij,t}^B \\ I_{ij,t}^B - U_{i,t} \end{array} \right\|_2 \leq I_{ij,t}^B + U_{i,t}, \forall ij \in \mathbf{L}, \forall t \in \mathbf{T} \quad (\text{B.6})$$

2) Polygon-based linearization

Similar to (22), (A.5) can be transformed using polygon-based linearization methods, as follows:

$$-\sqrt{3}(P_{i,t}^{\text{SD}} + S_i^S) \leq Q_{i,t}^{\text{SD}} \leq -\sqrt{3}(P_{i,t}^{\text{SD}} - S_i^S), \quad \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (\text{B.7})$$

$$-\frac{\sqrt{3}}{2} S_i^S \leq Q_{i,t}^{\text{SD}} \leq \frac{\sqrt{3}}{2} S_i^S, \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (\text{B.8})$$

$$\sqrt{3}(P_{i,t}^{\text{SD}} - S_i^S) \leq Q_{i,t}^{\text{SD}} \leq \sqrt{3}(P_{i,t}^{\text{SD}} + S_i^S), \forall i \in \mathbf{N}^{\text{VSC}}, \forall t \in \mathbf{T} \quad (\text{B.9})$$

Also, (23) can be transformed as:

$$-\sqrt{3}(P_{e,t}^D + S_e) \leq Q_{e,t}^D \leq -\sqrt{3}(P_{e,t}^D - S_e), \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (\text{B.10})$$

$$-\frac{\sqrt{3}}{2} S_e \leq Q_{e,t}^D \leq \frac{\sqrt{3}}{2} S_e, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (\text{B.11})$$

$$\sqrt{3}(P_{e,t}^D - S_e) \leq Q_{e,t}^D \leq \sqrt{3}(P_{e,t}^D + S_e), \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (\text{B.12})$$

3) Absolute value

Constraint (2) contains absolute-value functions, can be simplified by introducing binary variables, similar to (9), as follows:

$$d_{e,t} \geq \frac{SOC_{e,t} - SOC_{e,t-1}}{E_e}, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (\text{B.13})$$

$$d_{e,t} \geq \frac{SOC_{e,t-1} - SOC_{e,t}}{E_e}, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (\text{B.14})$$

$$d_{e,t} \leq \frac{SOC_{e,t-1} - SOC_{e,t}}{E_e} + (1 - x_e^{Abs})M, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (\text{B.15})$$

$$d_{e,t} \leq \frac{SOC_{e,t} - SOC_{e,t-1}}{E_e} + x_e^{Abs}M, \forall e \in \mathbf{E}, \forall t \in \mathbf{T} \quad (\text{B.16})$$

where x_e^{Abs} are the binary variables. If $SOC_{e,t-1} \geq SOC_{e,t}$, x_e^{Abs} is 1; otherwise, it is 0.

Also, (7) can be simplified as follows.

$$P_{i,t}^{loss} \geq (1 - \eta_{con}) \cdot P_{i,t}^{SD}, \forall i \in \mathbf{N}^{VSC}, \forall t \in \mathbf{T} \quad (\text{B.17})$$

$$P_{i,t}^{loss} \geq -(1 - \eta_{con}) \cdot P_{i,t}^{SD}, \forall i \in \mathbf{N}^{VSC}, \forall t \in \mathbf{T} \quad (\text{B.18})$$

$$P_{i,t}^{loss} \leq -(1 - \eta_{con}) \cdot P_{i,t}^{SD} + (1 - x_i^{Abs})M, \forall i \in \mathbf{N}^{VSC}, \forall t \in \mathbf{T} \quad (\text{B.19})$$

$$P_{i,t}^{loss} \leq (1 - \eta_{con}) \cdot P_{i,t}^{SD} + x_i^{Abs}M, \forall i \in \mathbf{N}^{VSC}, \forall t \in \mathbf{T} \quad (\text{B.20})$$

where x_i^{Abs} denotes binary variables. If $P_{i,t}^{SD} \leq 0$, x_i^{Abs} is 1, otherwise 0.

4) Rounding process

Constraint (5) contains rounding functions, can be simplified by introducing integer and binary variables, similar to (13), as follows:

The rounded value $\lceil \varepsilon_e \rceil$ can be represented by an integer variable n_e^R . Furthermore, the relationship between them should be

$$\varepsilon_e \leq n_e^R < \varepsilon_e + 1, \forall e \in \mathbf{E} \quad (\text{B.21})$$

In practice, strict inequality has to be approximated as

$$\varepsilon_e \leq n_e^R \leq \varepsilon_e + 1 - \varepsilon, \forall e \in \mathbf{E} \quad (\text{B.22})$$

where ε is a very small positive value.

Here, the integer variable n_e^R needs to be further simplified because the integerized values need to be multiplied by the continuous variables in (5). In the binary form, the integer variable n_e^R is represented by three binary variables as follows:

$$n_e^R = 4x_e^{R0} + 2x_e^{R1} + x_e^{R2}, \forall e \in \mathbf{E} \quad (\text{B.23})$$

$$C_{\text{IESS},e} = (4x_e^{R0} + 2x_e^{R1} + x_e^{R2}) \times (C_e^I E_e + C_p^I S_e S_B), \forall e \in \mathbf{E} \quad (\text{B.24})$$

where n_e^R are integer variables that can be combined with binary integer variables for a binary representation. where $n_e^R \in \{0, 1, \dots, 7\}$ denotes the number of ES in cycle Y .

The variables $y_e^{R0}, y_e^{R1}, y_e^{R2}, z_e^{R0}, z_e^{R1}, z_e^{R2}$ are introduced to simplify the product of the binary and continuous variables in (B.24).

$$\begin{cases} -M(1 - x_e^{R0}) \leq y_e^{R0} - 4E_e \leq M(1 - x_e^{R0}) \\ -Mx_e^{R0} \leq y_e^{R0} \leq Mx_e^{R0} \end{cases}, \forall e \in \mathbf{E} \quad (\text{B.25})$$

$$\begin{cases} -M(1 - x_e^{R1}) \leq y_e^{R1} - 2E_e \leq M(1 - x_e^{R1}) \\ -Mx_e^{R1} \leq y_e^{R1} \leq Mx_e^{R1} \end{cases}, \forall e \in \mathbf{E} \quad (\text{B.26})$$

$$\begin{cases} -M(1 - x_e^{R2}) \leq y_e^{R2} - E_e \leq M(1 - x_e^{R2}) \\ -Mx_e^{R2} \leq y_e^{R2} \leq Mx_e^{R2} \end{cases}, \forall e \in \mathbf{E} \quad (\text{B.27})$$

$$\begin{cases} -M(1 - x_e^{R0}) \leq z_e^{R0} - 4S_e \leq M(1 - x_e^{R0}) \\ -Mx_e^{R0} \leq z_e^{R0} \leq Mx_e^{R0} \end{cases}, \forall e \in \mathbf{E} \quad (\text{B.28})$$

$$\begin{cases} -M(1 - x_e^{R1}) \leq z_e^{R1} - 2S_e \leq M(1 - x_e^{R1}) \\ -Mx_e^{R1} \leq z_e^{R1} \leq Mx_e^{R1} \end{cases}, \forall e \in \mathbf{E} \quad (\text{B.29})$$

$$\begin{cases} -M(1 - x_e^{R2}) \leq z_e^{R2} - S_e \leq M(1 - x_e^{R2}) \\ -Mx_e^{R2} \leq z_e^{R2} \leq Mx_e^{R2} \end{cases}, \forall e \in \mathbf{E} \quad (\text{B.30})$$

Thus, a simplification of the rounding is obtained using (B.31).

$$C_{\text{IESS},e} = (4y_e^{R0} + 2y_e^{R1} + y_e^{R2}) \times C_e^I + (4z_e^{R0} + 2z_e^{R1} + z_e^{R2}) \times C_p^I S_B, \forall e \in \mathbf{E} \quad (\text{B.31})$$

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